Technical Report: Explainability System Design

German Credit Dataset

CSCE 4930: Ethical AI Project, Module 2

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1 Part 1: Model Training

For this module, we went with a **Decision Tree Classifier** capped at a maximum depth of 5. The reasoning here was pretty straightforward. Unlike the Random Forest we used in Module 1 (which is essentially a black box of hundreds of trees averaged together), a single shallow decision tree gives us something we can actually look at and follow step by step. You can literally trace the path from root to leaf for any given applicant and see exactly which features were checked and at what thresholds.

We reused the exact same preprocessing pipeline from Module 1, including the same train/test split, the same feature scaling, and the same one-hot encoding. The only thing that changed is the classifier itself.

Performance

The model's performance metrics are shown in Table 1.

Table 1: Decision Tree Performance Metrics

Metric	**Value**
Accuracy	0.685
Precision	0.725
Recall	0.886
F1-Score	0.797
Tree Depth	5
Leaves	28
Features Used	21 / 65

These numbers are lower than the Random Forest baseline from Module 1 (which had 0.735 accuracy). That is expected because we deliberately traded raw predictive power for interpretability. A shallow tree with only 5 levels and 28 leaves simply cannot capture

all the complex interactions a 100-tree ensemble can. Out of the 65 features produced by preprocessing, the tree only ended up using 21 of them (Table 1). But the whole point of this module is understanding *why* the model decides what it decides, and that trade-off is worthwhile here.

The classification report breaks this down further by class. The model achieves 0.886 recall for "Good Credit" but only 0.22 recall for "Bad Credit," meaning it correctly identifies most good applicants but misses a lot of bad ones. This skew makes sense given the class imbalance in the dataset (700 good vs. 300 bad in the full set).

2 Part 2: Why Is This Model Interpretable?

A decision tree is interpretable *by design* because it mirrors how humans naturally make decisions: a sequence of yes/no questions.

Think of it this way. When a loan officer reviews an application, they might mentally go through something like: "Is the checking account balance above a certain threshold? If yes, how long is the loan duration? If it's short, approve; if it's long, check the credit history..." A decision tree does exactly this, but in a formalized and reproducible manner.

There are a few specific reasons why this transparency matters:

1. **Each prediction has a traceable path.** For any individual, we can follow the exact sequence of feature checks (splits) from the root node down to the leaf. There is no hidden layer and no weighted sum of 500 trees. It is just a single path of "if-then" logic. This is demonstrated concretely in the Part 5 local explanations (see Figure 4).

2. **Feature importance is built in.** The tree tells us directly which features it relies on most, measured by how much each feature reduces impurity (Gini index) across all the splits where it appears. This is what we plot in Figure 1.

3. **The max depth constraint keeps it human-readable.** With `max_depth=5`, no decision path is longer than 5 questions. A person can realistically hold 5 conditions in their head and verify whether the logic makes sense. If we allowed depth 20, the tree would be accurate but incomprehensible.

4. **Thresholds are explicit and auditable.** Each split says something concrete like "duration $\leq$ 22.5 months." A domain expert can look at that and judge whether the threshold is reasonable. Compare that with a neural network weight of 0.0347, which means nothing in isolation.

In short, this model is interpretable because every part of its decision-making is laid bare. You do not need any external explanation tool (like SHAP or LIME) to understand it. The model *is* the explanation.

3 Part 3: What Features Matter the Most?

The feature importance analysis is captured in Table 2 and Figure 1. Feature importance here is measured by Gini importance: the total reduction in node impurity contributed by each feature across all splits in the tree.

Table 2: Top 10 Most Important Features (Gini Importance)

Rank	Feature	Importance
1	status_no checking account	0.311
2	duration	0.176
3	amount	0.101
4	other_debtors_guarantor	0.057
5	savings_unknown/no savings account	0.054
6	other_installment_plans_none	0.039
7	purpose_car (used)	0.033
8	property_real estate	0.025
9	age	0.024
10	credit_history_critical account	0.024

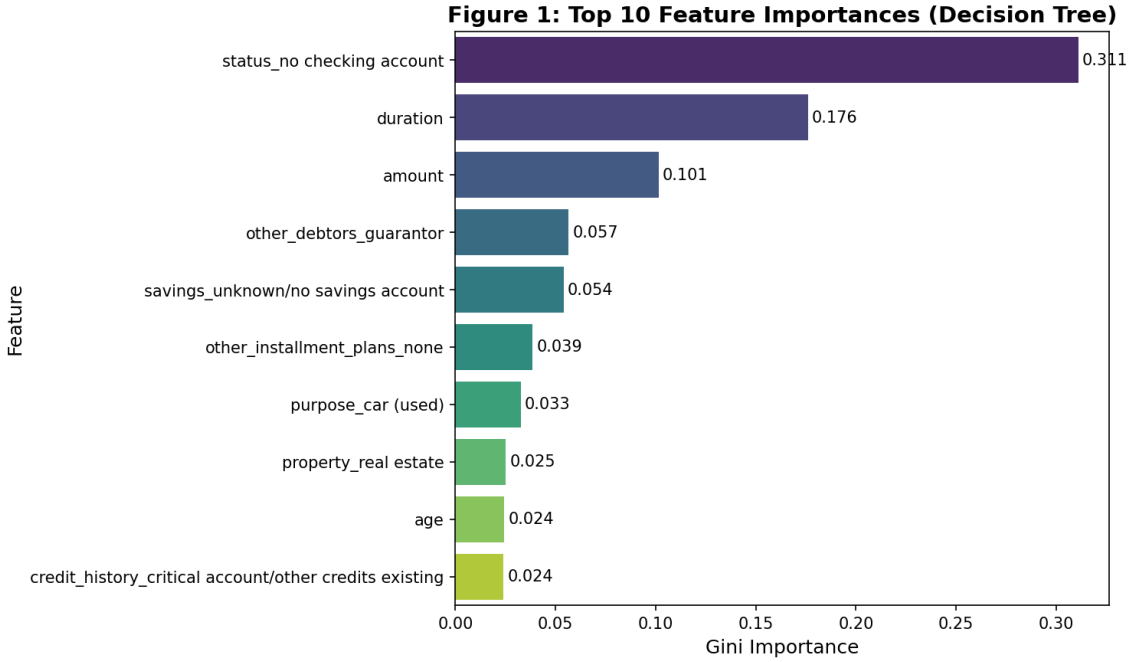

Figure 1: Top 10 Feature Importances from the Decision Tree, ranked by Gini importance.

Do they make sense?

Largely, yes. The single most important feature, whether the applicant has no checking account, makes strong intuitive sense. In the German credit system, not having a checking account is a significant negative signal; it suggests limited engagement with the formal banking system. The tree uses this as its very first split (the root node), which means it is the single most informative piece of information for separating good from bad credit risks.

Loan duration and credit amount ranking second and third are equally intuitive. Longer loans and larger amounts carry more risk, all else being equal. Having a guarantor (ranked 4th) provides a safety net that reduces default risk. Credit history and savings round out the top features as expected financial indicators.

However, `age` appearing at rank 9 with importance 0.024 is a concern. Age is a protected attribute, and its presence in the tree means the model is directly using demographic information to make predictions. We investigate this further in Part 6.

Also notable: 44 out of 65 features ended up with zero importance. The tree performs implicit feature selection, meaning it only splits on what actually helps separate the classes. Most one-hot encoded categories turned out to be irrelevant.

4 Part 4: How Do Features Affect Predictions?

This is where we move from "which features matter" to "how do they actually influence the outcome." The analysis is captured in Table 3 (Split Analysis), Figure 2 (tree visualization), and Figure 3 (split effect chart).

For each feature that appears in the tree, we examined:

- The **threshold** at which the split occurs
- Which **direction** (above or below the threshold) leads to a higher probability of being classified as "Good Credit"

Table 3: Key Decision Tree Splits and Their Directional Effects

Feature	**Threshold**	**Direction Favoring Good Credit**
`status_no checking account`	0.50	> 0.50 (no checking acct.)
`duration`	-0.77	≤ -0.77 (shorter loans)
`amount`	-0.65	> -0.65 (moderate amounts)
`age`	-0.25	> -0.25 (older applicants)
`other_debtors_guarantor`	0.50	> 0.50 (has guarantor)
`savings_unknown/no savings`	0.50	> 0.50 (unknown savings)
`other_installment_plans_none`	0.50	> 0.50 (no other plans)
`property_real estate`	0.50	> 0.50 (owns real estate)
`credit_history_critical`	0.50	> 0.50 (critical history)
`personal_status_sex_female`	0.50	≤ 0.50 (not female)
`sex_male`	0.50	> 0.50 (is male)

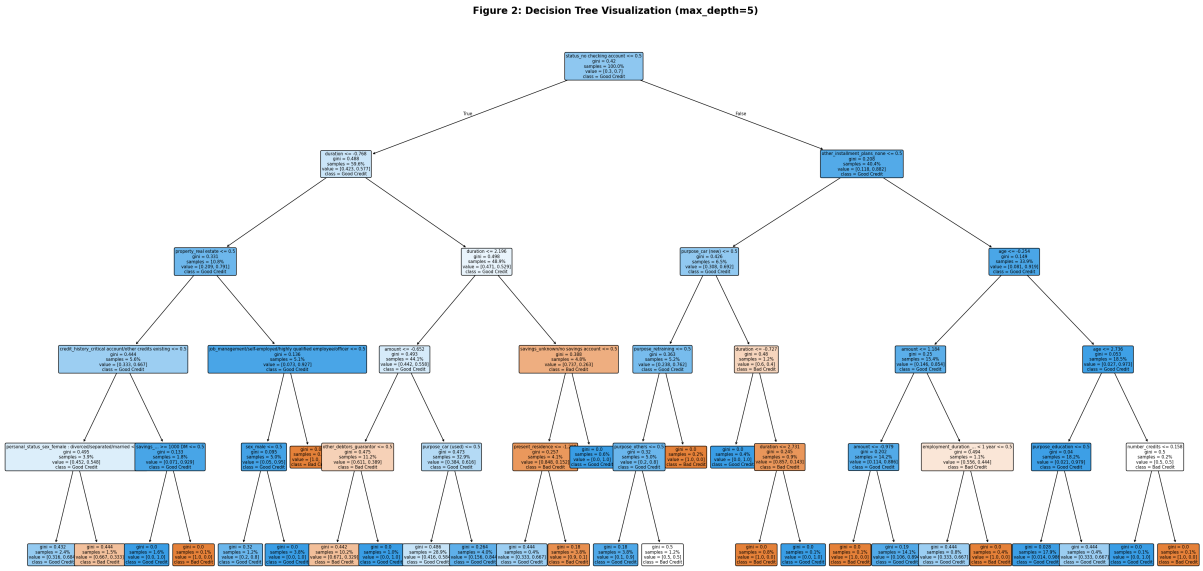

Figure 2: Full Decision Tree visualization (max_depth=5). Nodes are colored by majority class: darker shading indicates higher confidence.

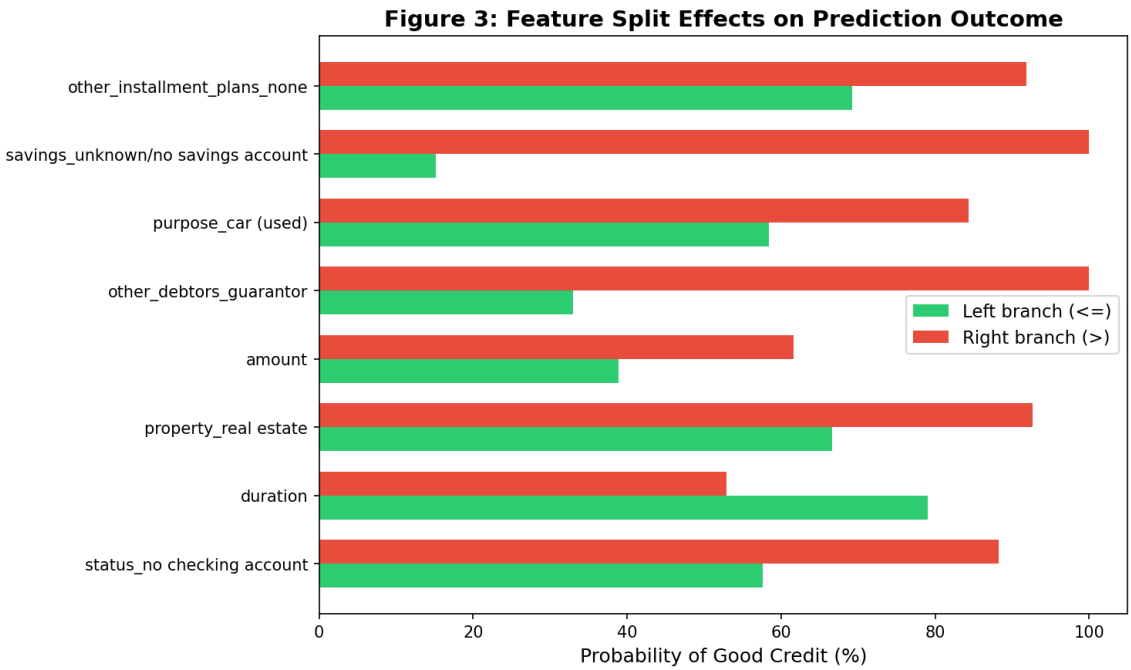

Figure 3: Feature split effects on prediction outcome. Green bars show the probability of Good Credit in the left branch ($\leq$ threshold); red bars show the right branch ($>$ threshold).

The results from Table 3 and the tree rules are revealing. The root split on `status_no checking account` > 0.50 immediately favors Good Credit. Applicants with no checking account on file actually get routed to a more favorable branch. This might seem counter-intuitive at first, but in the German Credit dataset "no checking account" often means

the applicant has no overdraft risk visible to the bank, which paradoxically correlates with better outcomes.

For `duration`, shorter loans (scaled value ≤ -0.77) favor Good Credit, which is straightforward since less time means less risk. But the tree also splits on duration again deeper down at threshold 2.20, creating a non-monotonic effect: very long loans are risky, moderate ones are fine, and short ones are best.

The feature `age` splits at -0.25 (standardized), with values above this threshold favoring Good Credit. In raw terms, this means older applicants get better outcomes. The tree also splits on `personal_status_sex_female`, where being in that category (> 0.50) *hurts* the prediction. Being female pushes the model toward Bad Credit at that particular branch.

This non-monotonic and context-dependent behavior is one of the key differences between trees and linear models. Unlike logistic regression, where each feature has a single fixed coefficient, the tree can use the same feature at multiple levels with different thresholds and different effects depending on what path was taken above. Table 3 captures the primary split for each key feature.

5 Part 5: Local Explanations (2 Individuals)

We selected two test-set individuals: one who was correctly classified as **Good Credit** and one correctly classified as **Bad Credit**. The full decision paths are printed in the notebook output, and the feature value comparison is shown in Figure 4.

Individual 1: Approved (Good Credit)

This is a 42-year-old male with a loan duration of 18 months and an amount of 2,427 DM. The model predicted Good Credit with a confidence of 58.4% (Bad=41.6%, Good=58.4%). The prediction matched the actual label.

The decision path through the tree (5 steps, as printed in the notebook):

1. `status_no checking account` $= 0.0 \leq 0.5 \rightarrow$ go LEFT (has a checking account)
2. `duration` $= -0.233 > -0.768 \rightarrow$ go RIGHT (not a short loan)
3. `duration` $= -0.233 \leq 2.196 \rightarrow$ go LEFT (not an extremely long loan either)
4. `amount` $= -0.274 > -0.652 \rightarrow$ go RIGHT (moderate loan amount)
5. `purpose_car (used)` $= 0.0 \leq 0.5 \rightarrow$ go LEFT (not for a used car)
6. Reaches a **leaf**: Good Credit (58.4% Good)

In plain language: this person had a checking account, a moderate-length and moderate-amount loan that was not for a used car, and the model approved them. The confidence is not overwhelming (58%), which reflects the fact that this combination of features is not a clear-cut approval. It is a borderline case that tipped favorable.

Individual 2: Denied (Bad Credit)

This is a 33-year-old male with the same loan duration of 18 months but a lower amount of 1,245 DM. Despite the smaller loan, the model predicted Bad Credit with 67.1% confidence. The prediction was correct.

The decision path diverged from Individual 1 at step 4:

1. `status_no checking account` $= 0.0 \leq 0.5 \rightarrow$ go LEFT (same as Individual 1)
2. `duration` $= -0.233 > -0.768 \rightarrow$ go RIGHT (same)
3. `duration` $= -0.233 \leq 2.196 \rightarrow$ go LEFT (same)
4. `amount` $= -0.701 \leq -0.652 \rightarrow$ go LEFT (**different**: lower amount)
5. `other_debtors_guarantor` $= 0.0 \leq 0.5 \rightarrow$ go LEFT (no guarantor)
6. Reaches a **leaf**: Bad Credit (67.1% Bad)

This is a striking result. The two individuals followed the same path for the first three splits but diverged because Individual 2's loan amount was *lower*. Counterintuitively, the smaller loan pushed them into a riskier branch. The likely explanation is that among applicants with checking accounts and moderate-duration loans, a very small credit amount is associated with financially precarious borrowers. Then, the lack of a guarantor sealed the denial.

Comparison

Figure 4 shows the scaled feature values for both individuals side by side, limited to the features the tree actually uses. The critical difference turned out to be just two features: credit amount and having a guarantor. Everything else was nearly identical.

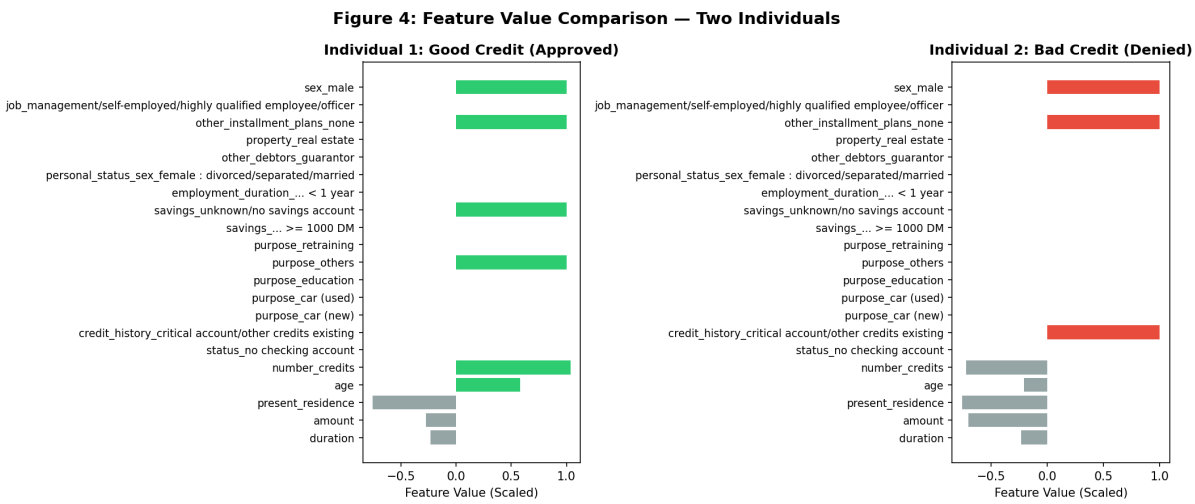

Figure 4: Feature value comparison for the two explained individuals. Left: Individual 1 (approved). Right: Individual 2 (denied). Only features used by the tree are shown.

This comparison highlights something important about decision trees: the decision boundary is sharp. A small change in one feature can flip the outcome entirely. Individual 2 might have been approved if they had a guarantor or if their loan amount was slightly higher. That kind of concrete, actionable feedback ("get a guarantor and reapply") is exactly the transparency that makes decision trees valuable in regulated settings.

6 Part 6: Detect Suspicious Patterns

This is arguably the most important section from an ethical AI perspective. We ran four checks, with results visualized in Figure 5 and Figure 6.

Check 1: Direct use of protected attributes

The tree **does** directly use protected attributes. Check 1 in the notebook flagged the following features with non-zero importance:

- `age`: importance 0.024
- `personal_status_sex_female : divorced/separated/married`: importance 0.016
- `sex_male`: importance 0.005

This means the tree is explicitly splitting on age and sex-related features. That is a clear problem for fair lending. A loan decision that hinges partly on whether someone is female or how old they are is ethically and legally questionable.

Check 2: Prediction rate disparities

The disparities are significant, as shown in Table 4 and Figure 5:

Table 4: Prediction and Error Rates by Demographic Group

Attribute	**Group**	**Pred. Rate**	**Error Rate**	**n**
Age Group	age_25_and_over	0.866	0.285	172
	under_25	0.786	0.500	28
Sex	male	0.900	0.286	140
	female	0.750	0.383	60

The model is essentially wrong half the time for young applicants (50% error rate vs. 28.5% for older ones). Males get approved at 90% while females only reach 75%. These gaps are substantial.

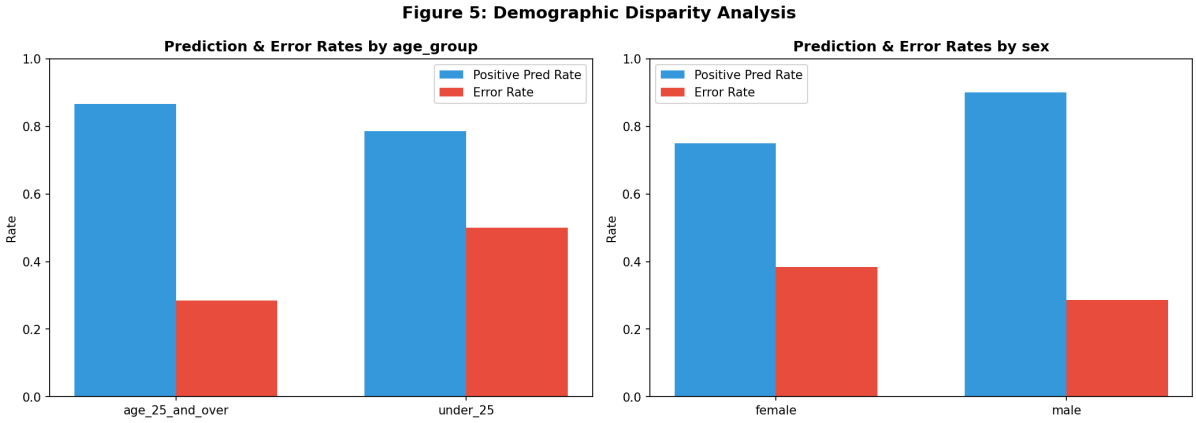

Figure 5: Prediction rates and error rates by age group and sex. The gaps between groups indicate demographic disparities in model behavior.

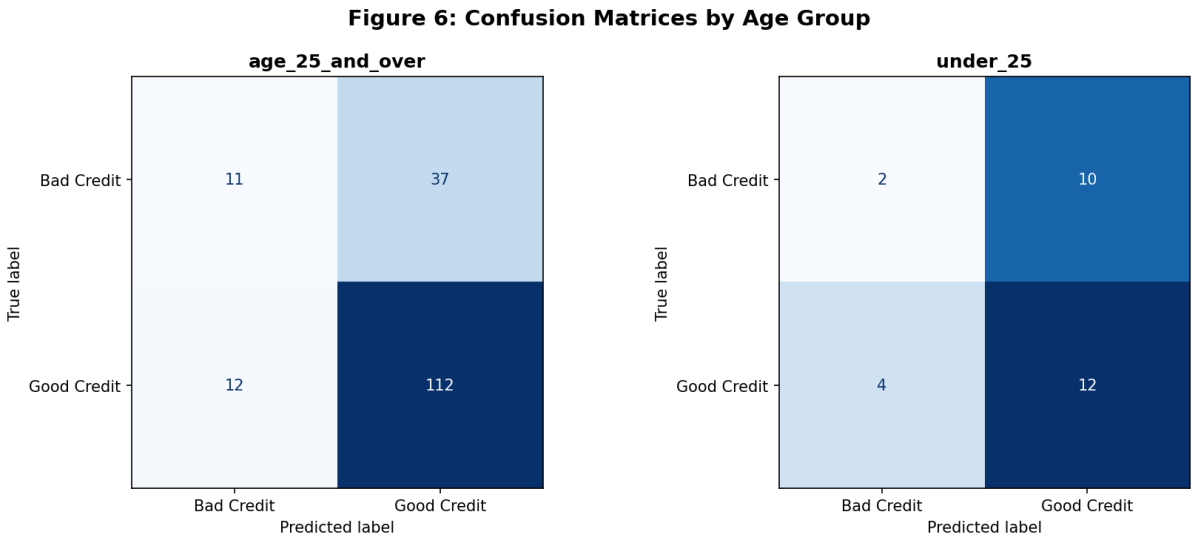

Figure 6: Confusion matrices by age group. These reveal the specific types of errors (false positives vs. false negatives) for each demographic group.

The confusion matrices by age group (Figure 6) break this down further. They reveal that the high error rate for younger applicants is likely driven by a high false positive or false negative rate, meaning the model is systematically miscalibrated for this group.

Check 3: Importance of sensitive features

In total, sensitive features account for **6.0% of the tree's total importance** (0.060 out of 1.0). While this is not the majority of the model's decision-making, it is far from negligible. The `age` feature alone (0.024) ranks 9th out of 65 features, which is higher than many financial indicators. The tree also splits on `job_management/self-employed` (importance 0.015), which was flagged because job type can correlate with age.

Check 4: Proxy detection via correlation

The proxy correlation analysis revealed one clear flag: the `age` feature itself has a correlation of **0.485** with the `age_group` protected attribute. This is expected (age *is* the protected attribute), but it confirms the tree is using it.

Notably, none of the other top features showed correlations above 0.3 with either age or sex. This means the bias is not hiding behind proxies. It is out in the open, directly encoded in the tree's splits on `age`, `sex_male`, and `personal_status_sex_female`.

Connection to Module 1

In Module 1, we found that the baseline Random Forest had a Demographic Parity Difference of 0.147 and an Equalized Odds Difference of 0.224 with respect to age group. The current Decision Tree shows a prediction rate gap of about 8 percentage points between age groups (86.6% vs. 78.6%), which is comparable in magnitude. The sex disparity (90.0% vs. 75.0%, a 15-point gap) is arguably worse. So switching to an interpretable model did not fix the fairness problem, but it made the bias *visible and traceable*, which is the whole point of this module.

7 Part 7: Design Recommendation

Would we deploy this model?

Honestly, not as-is. While this Decision Tree is a solid step toward transparency, there are a few reasons we would want to iterate further before real deployment:

1. **Accuracy trade-off.** At 68.5% accuracy (Table 1), the tree is 5 percentage points below the Module 1 Random Forest (73.5%). In a credit-scoring context, that gap translates directly into money lost (from approving bad risks) or opportunities denied (from rejecting good applicants). The recall for Bad Credit is particularly poor at just 22%, meaning the model misses nearly 4 out of 5 genuinely bad risks. A production system would need to hit a higher bar to be commercially viable.

2. **Direct use of protected attributes.** As discussed in Part 6, the tree does not just use proxies. It directly splits on `age`, `sex_male`, and `personal_status_sex_female`. These features account for 6% of total importance (Table 2). Before deployment, we would need to remove these features entirely, retrain, and verify that the accuracy loss is acceptable. Then apply fairness constraints similar to what we did in Module 1.

3. **Robustness.** A single decision tree is notoriously sensitive to small changes in the training data. Removing a handful of samples can completely change the tree structure. For deployment, we might want to consider a small ensemble (like a depth-limited Random Forest with 5 to 10 trees) that preserves some interpretability while gaining stability.

What are the risks?

The primary risk is **allocative harm**, meaning the systematic denial of credit to qualified applicants from certain demographic groups. The data already shows this happening

(Table 4). Females get approved at 75.0% vs. males at 90.0%, and young applicants face a 50% error rate. If the underlying patterns in the training data reflect historical discrimination, the model will learn and reproduce those patterns.

A secondary risk is **over-trust in interpretability**. Because this model can be explained, there is a temptation to assume it is fair. But interpretability does not imply fairness. A perfectly interpretable model can still be perfectly biased; the bias is just easier to spot.

Finally, there is a **gaming risk**. If the tree structure becomes public (Figure 2), applicants could manipulate the exact features the tree checks to get approved. With only 5 levels of splits, the set of "rules to pass" is relatively small and exploitable.

What would we change?

If we were moving toward production, we would:

- Apply the fairness mitigation techniques from Module 1 (resampling or Exponentiated Gradient) to this interpretable model
- Remove or transform proxy features identified in Part 6 before retraining
- Validate on a held-out temporal dataset (not just a random split) to check for distribution shift
- Pair the model with a human review process for borderline cases, especially for applicants from underprivileged groups
- Run a formal disparate impact analysis to comply with regulatory requirements (e.g., the Equal Credit Opportunity Act)

The interpretable tree is a strong *starting point* for a fair and transparent credit scoring system, but it needs the fairness guardrails and operational safeguards layered on top before it is ready for the real world.